Nicholas Johnson

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EDUCATION

George Mason University Expected Summer 2027

B.S. Neuroscience, Concentration: Computational Neuroscience | Fairfax, VA

- Curriculum focused in cellular neuroscience, computational neuroscience, and neuroinformatics.
- Selected for GEO Neurotechnology in Germany Program, a 6-credit program visiting 8 research labs across Germany.
- Treasurer, Students in Neuroscience; administering Fall 2026 budget and expense tracking.

RESEARCH

NeuroMorpho.org Research Assistant June 2026–Present

Neuroinformatics, Krasnow Institute for Advanced Study, George Mason University | Fairfax, VA

Literature Mining

- Conduct literature mining to identify neuronal and glial reconstructions for NeuroMorpho.Org database curation.
- Evaluate reconstruction evidence using tracing-system keywords including Neurolucida, Imaris/Filament Tracer, Amira, Vaa3D, Simple Neurite Tracer, and CATMAID.
- Curate morphology metadata by verifying article details, reconstruction counts, figure evidence, supplementary files, data availability, and DOI/PMID references.
- Review repositories including GitHub, Dryad, Figshare, Zenodo, and Dataverse for downloadable reconstruction data.

Neuron Morphology Editing

- Mark somas and dendrites to correctly classify reconstruction components.
- Convert multi-point somas into single-point soma representations.
- Fix erroneous long connections between reconstruction nodes.
- Repair branch structure to preserve accurate neuronal topology.

Undergraduate Research Assistant January 2026–May 2026

Anatomy and Physiology Education Research, George Mason University | Fairfax, VA

- Collaborated on manuscript preparation for a comparative anatomy and physiology education study.
- Contributed to data interpretation and formation of graphs in google colab for peer-reviewed publication.

PROJECTS

Hodgkin–Huxley Simulator May 2026–Present

Independent Project | hhsimulator.com

- Built a cross-platform desktop application (Windows, macOS, Linux) implementing the Hodgkin–Huxley equations to model action potential generation and membrane dynamics in real time.
- Simulates single-compartment membrane dynamics with live visualization of membrane voltage, injected current, net ionic current, conductances (Na^+, K^+, leak), and gating variables (m, h, n).
- Validated outputs against the NEURON Simulator; authored a research-style manuscript covering methods, error analysis, biological interpretation, and future extensions.

WORK EXPERIENCE

AI Trainer / LLM Evaluator 2026–Present

Handshake AI Fellowship | Remote

- Evaluated large language model responses for accuracy, reasoning quality, instruction-following, safety, and overall usefulness.
- Provided structured feedback on prompts and outputs to help improve AI assistant behavior across varied tasks and domains.

MRI Office Assistant July 2025–Present

Institute for Biohealth Innovation, George Mason University | Fairfax, VA

- Support research administration office operations and completed entry-level MRI training from a certified MRI Technologist.

Office Assistant, IET Department February 2025–May 2025

Northern Virginia Community College | Annandale Campus, VA

- Built computational computer-network labs, supported hardware setup, laptop debugging, and switch configuration for NOVA students using AI supported tools.

TECHNICAL SKILLS

Programming & Tools	Python, GitHub, Google Colab
Modeling & Simulation	NEURON Simulator, Hodgkin–Huxley modeling, single-compartment membrane dynamics
Neuroinformatics	NeuroMorpho.Org curation, cvapp, morphology editing, metadata verification, literature mining
AI	Prompt development, LLM output review, factual accuracy assessment, reasoning quality assessment
Cellular Neuroscience	Electrical properties of nerve cells, synaptic transmission, neurotransmitters and receptors, molecular signaling, synaptic plasticity